

### Assignment – 1

| Subject Name & Code                             | Semester        | Issue Date | Due Date   |
|-------------------------------------------------|-----------------|------------|------------|
| Object Oriented Design & Programming / 01AI0103 | 2 <sup>nd</sup> | 06-02-2025 | 25-02-2025 |

| Sr. No. | Question                                                                                                                                     | Course Outcome |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1       | What is Object Oriented Programming? Write any five features of OOP.                                                                         | CO1            |
| 2       | Differentiate between C and C++ programming language.                                                                                        | CO1            |
| 3       | Explain various Access modifiers for C++ classes.                                                                                            | CO1            |
| 4       | Describe the following Object Oriented Programming Concepts:<br>i)Encapsulation<br>ii)Data Abstraction<br>iii)Polymorphism<br>iv)Inheritance | CO1            |
| 5       | What is type conversion? Explain the two types of type conversion in C++ with example.                                                       | CO1            |
| 6       | Describe the use of following manipulators with syntax:<br>setw, setfill, setbase, endl                                                      | CO2            |
| 7       | Explain Call by Value and Call by Reference with appropriate example.                                                                        | CO2            |
| 8       | What is friend function? Explain with example.                                                                                               | CO2            |
| 9       | What is inline function? Explain with example                                                                                                | CO2            |
| 10      | Write a program to perform Function Overloading.                                                                                             | CO2            |
| 11      | What are Macros Function? Give example of function-like macros.                                                                              | CO2            |
| 12      | Write short notes on :<br>i. User-Defined Functions<br>ii. Default arguments                                                                 | CO2            |
| 13      | Explain the role of parameterised and copy constructor with appropriate example.                                                             | CO2            |
| 14      | Explain the use of destructor with example.                                                                                                  | CO2            |
| 15      | What is a class and object? Write a simple program using class in C++ to input subject mark and prints it.                                   | CO2            |